DATUM / ENGINEERING OS

Robot Arm V2

Robot-arm design, continuous-torque analysis, and the evidence still needed for hardware validation.

Generated project report · evidence basis and limitations retained

01 PROJECT A six-function desktop robot

BRIEF

arm, designed as a system

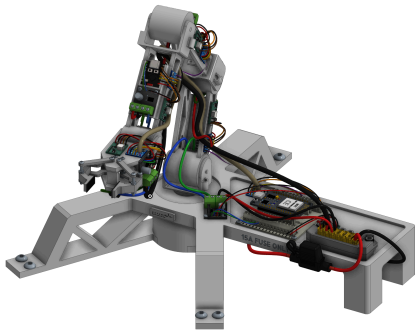
RoboArm V2 brings mechanics, actuation, controls, power, motion analysis, and evidence into one virtual prototype.

6	12.728 in	1.50 lbf
actuated functions	shoulder-to-jaw reach	rated design payload

The current design uses two shoulder stages, an elbow, wrist, rotating base, and powered claw. The official full-extension case is documented, but physical capacity has not been demonstrated.

Virtual prototype

Analysis supports feasibility. It does not yet prove manufactured-hardware performance.



Integrated CAD assembly · source-backed render

PORT-05 Source claim

RoboArm V2 system-level project summary

RoboArm V2 is documented as a system-level virtual design of a six-function desktop robotic

DEC-04 Source claim

Calculation revision hierarchy

The current project definition designates the 12.728 in geometry, 1.50 lbf official payload, current FAULHABER candidates, and latest

arm with 12.728 in reach and an official 1.50 lbf full-extension payload. The project integrates mechanical CAD, actuator selection, static/dynamic torque analysis, distributed ESP32/CAN controls, 12 V power distribution, routed wiring, and an N20-driven claw. It remains a virtual prototype requiring final component analysis, thermal/electrical verification, detailed joint hardware, and physical testing.

Recorded Inspect evidence ↗

joint-by-joint PDF torque study for formal documentation. The early all-N20 sizing, 27.82688 lbf·in shoulder walkthrough, and earlier claw-mount FEA remain historical/superseded process evidence.

Recorded Inspect evidence ↗

02 SCOPE + ARCHITECTURE

What the project covers

The design scope is system-level: structure, four primary torque checks, base motion, a powered end effector, distributed controls, and 12 V power.

01	02	03	04	05	06
Base rotation	Lower shoulder	Upper shoulder	Elbow	Wrist	Claw

REQUIREMENT COVERAGE

"Recorded" means documented in the project—not physically verified.

REQ-01	Official payload rating	Recorded
REQ-02	Continuous joint torque criterion	Needs review
REQ-03	Reach and reference configuration	Recorded
REQ-04	Motion envelope and accuracy	Open
REQ-05	Electrical protection and safe stopping	Needs review

REQ-06	Structure and gripper acceptance	Open
REQ-07	System actuation architecture	Recorded

CMP-03 Source claim

Distributed controller and CAN architecture

Distributed joint control over CAN; exact electronics BOM and interfaces TBD

Needs review Inspect evidence

CMP-04 Source claim

12 V power distribution and protection

12 V shared bus; 14 AWG primary wiring stated; 15 A modeled fuse label

Needs review Inspect evidence

CMP-05 Source claim

N20 claw and 25:10 belt reduction

N20 drive; 25T:10T / 2.5:1 transmission; final grip force TBD

Needs review Inspect evidence

03 ANALYTICAL PERFORMANCE

Torque feasibility across four loaded joints

The current model compares continuous output limits with Ku-adjusted gravity and inertia demand at the official payload.

All four modeled ratios exceed 1.0×.

These are actuator-model comparisons. They are not structural safety factors, certified capacity, or thermal-duty validation.

Joint	Demand	Claimed limit	Ratio
Lower shoulder	58.43 lbf·in	88.5 lbf·in	**1.51×**
Upper shoulder	39.90 lbf·in	88.5 lbf·in	**2.22×**
Elbow	21.22 lbf·in	32.6 lbf·in	**1.54×**
Wrist	7.74 lbf·in	32.6 lbf·in	**4.21×**

CALC-06 Derived

Current full-extension static shoulder case

From the current project summary: payload moment 19.092 lbf·in plus approximately 12.0 lbf·in arm self-weight moment gives approximately 31.092 lbf·in required static shoulder torque. Compared with approximately 52 lbf·in continuous output, the stated margin is approximately 1.7×. The separate structural FEA factor cited as approximately 2.0 is not this actuator margin.

Analyzed Inspect evidence ↗

RISK-01 Missing

Actuator ratings and duty limits need source verification

Exact FAULHABER motor and gearbox candidates are now documented, closing the identity gap. The supplied evidence still lacks manufacturer curves/data for the cited geared-output limits at intended voltage, speed, controller, temperature, efficiency, and duty. Motor-shaft, geared-output, peak/stall, and sustainable continuous ratings remain non-interchangeable.

Open Inspect evidence ↗

04 ASSUMPTIONS + DECISIONS

The choices underneath the model

These statements make the analysis legible. Assumptions remain assumptions until

their applicability is confirmed.

01

20% torque uncertainty multiplier

Uploaded report explicitly applies Ku = 1.20 to gravity plus inertia torque; it does not independently quantify all uncertainty.

Ku = 1.20

02

Combined peak gravity and acceleration

Report combines maximum gravity with peak acceleration in a reduced model. Full coupled dynamics, friction, impact, and efficiency are not established.

Magnitude envelope for the stated model

03

Shared local-Y shoulder channel

Report shares the local-Y channel for both shoulders and treats 2.942 lbm as the complete downstream rotating assembly.

Shared α = 2.4264 rad/s²; assembly 2.942 lbm

REVISION + SCOPE DECISIONS

Current and historical cases stay separated.

DEC-04 Source claim

Calculation revision hierarchy

The current project definition designates the 12.728 in geometry, 1.50 lbf official payload, current FAULHABER candidates, and latest joint-by-joint PDF torque study for formal documentation. The early all-N20 sizing, 27.82688 lbf·in shoulder walkthrough, and earlier claw-mount FEA remain

DEC-05 Source claim

Six-function system scope

V2 is documented as a six-function serial articulated system: base, lower shoulder, upper shoulder, elbow, wrist, and claw. Base yaw remains outside the shoulder-supported-mass torque model because its actuator is below the shoulder.

historical/superseded process evidence.

Recorded Inspect evidence ↗

Recorded Inspect evidence ↗

05 EXCLUSIONS + OPEN WORK

What this report does not claim

The most important portfolio detail is the boundary between an analyzed design and a validated machine.

DELIBERATE MODEL EXCLUSIONS

Base-yaw torque is outside the four-joint shoulder-supported-mass model.

Full coupled dynamics, friction, impact, drivetrain efficiency, and thermal behavior are not established.

Historical all-N20 sizing, the 27.82688 lbf·in shoulder case, and earlier claw-mount FEA are excluded from the current final case.

NOT YET VALIDATED

No manufactured prototype or physical payload test.

No current-V2 structural, fatigue, bearing, shaft, fastener, or anchoring validation.

No verified electrical, thermal, communications, stopping, or power-loss behavior.

No verified grip force, retention, repeatability, accuracy, clearance, or cycle time.

OPEN RISK REGISTER

7 open items keep the limits visible.

RISK-01	**Actuator ratings and duty limits need source verification** Verify rating basis before physical capacity or thermal-duty claims	Open
RISK-03	**Motion channel / frame ambiguity** Resolve object, axis, interval, fps	Open
RISK-04	**Rendered motion is not a hardware test** No physical pass recorded	Open

RISK-05	**Electrical sizing, communication, and safe stopping** Electrical, communications, thermal, and fault review pending	Open
RISK-06	**Current-V2 strength and payload retention unverified** Current V2 structure / retention review pending	Open
RISK-07	**Actuator envelopes exceed nominal joint housings** Exact motors, gearboxes, encoders, mounts, and cables integrated without interference	Open
RISK-08	**Base yaw and feedback definition incomplete**	Open

06 EVIDENCE + NEXT STEPS

A traceable path from design to proof

The portfolio can show the design process now. Stronger performance claims wait for manufacturer data, current-V2 analysis, and controlled testing.

14	5	7	0
retained source files	portfolio evidence records	open risks	verified physical tests

RECOMMENDED VERIFICATION SEQUENCE

Turn feasibility into evidence

Each step opens the live project record, where acceptance limits and evidence can be revised with history intact.

01 **Reproduce motion-to-torque analysis**
Numeric agreement and approved channel map

02 **Verify CAD properties, reach, and actuator packaging**
Traceable mass properties, 12.728 in reach definition, and interference-free actuator integration

03 **Power, controls, and communication bench review**
Approved schematic, current budget, interface map, and fault criteria

04 **Current-V2 structure and gripper verification**
Approved stress, deflection, life, belt-load, grip-force, and retention limits

05 **Controlled payload hold and motion**
Intended 1.500 lbf case; conditions TBD

06 **Travel, repeatability, and clearance**
Travel / accuracy targets TBD

Presentable now—with its qualification intact.

Describe RoboArm V2 as a detailed system-level virtual design and feasibility study, not a manufactured or validated product.